

JASNOOR SANDHU

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EDUCATION

Indian Institute of Technology (BHU)	2025, Varanasi
- B.Tech. major in Electronics and Communication Engineering (4 year), CGPA: 9.50/10 - Minor Degree in Computer Science Engineering - Department Rank 3/146 in academic batch	
Sudesh Vatika Convent School	2021, Talwandi Sabo
Scored First Grade in Senior Secondary Schooling (91.6%)	
St. Joseph's High School	2019, Mandi Dabwali
Scored First Grade in High School (92.5%)	

EXPERIENCE

Summer@EPFL Intern	Summer 2025, Lausanne Switzerland
- Improved the buffer placement algorithm in the Dynamatic HLS compiler (EPFL LAP), under Prof. Paolo lenne, focusing on MILP-based scheduling for dataflow circuits. - Identified and corrected missing throughput constraints, improving accuracy of buffer insertion.	
Hardware Intern at Siemens EDA	Summer 2024, Noida India
- Revamped a utility to detect faulty flip-flops caused by compiler issues or emulation errors. - Established a strong understanding of industry-standard codebases and architectures, also enhancing the utility by incorporating user-friendly features, like progress reports, flop filtering and waveform visualisation. - Developed advanced Linux Shell & C++ skills to improve the Flip Flop Verification tool, reducing analysis time by 50% and enhancing service efficiency, now used by multiple team members.	
Research Intern at Victoria University	Summer 2023, Melbourne Australia
- Modified ROS2 packages for Turtlebot4 navigation and mapping, gaining proficiency in Raspberry Pi4, and working in Linux environments. - Organized demonstration of Turtlebot4 for 40+ students at university.	

SKILLS

Programming Languages: C/C++, Python, Verilog/SystemVerilog, Linux Shell, Perl
Familiar with: OOPS, Linux, Data Structures
Software: MATLAB, Xilinx Vivado/Vitis, Proteus
Interests: RTL & Digital Design, FPGA, ASIC

PROJECTS

Goldsmith Division Algorithm IP Development	August, 2024
- Designed and implemented an IP block for the Goldsmith Division Algorithm using AXI Stream. - Successfully integrated and tested the IP on the Pynq Z1 FPGA board, achieving floating-point divisions in just 6 cycles. - Achieved 99.99% accuracy in floating-point division, demonstrating high precision and efficiency in FPGA implementation.	
Waveform Design for next-generation Wireless Communication	January-March, 2024
- Investigated the transition from OFDM to OTFS, enhance spectral efficiency by 20% and reliability for mobile broadband and IoT applications. - Performed simulations to evaluate BER performance under various signal-to-noise ratio conditions, demonstrating significant improvement of 30% in low SNR scenarios.	
RTL Implementation of Bubble Sort Algorithm	August 2023
- Created data flow model and state diagram for Bubble Sort Algorithm. - Learned about sorting algorithms, Verilog, RTL Design, and Vivado.	

EXTRA CURRICULAR AND ACHIEVEMENTS

Joint General Secretary of Electronics Engineering Society (2024)
• Managed Coordination among Marketing, Design, Web, Publicity and PR Teams for EES Fest'24 • Successfully organized and completed the EES Fest, attracting a total footfall of 800 students
Event Manager at Technex'24 (2024)
• Organised RoboWars Event, managing 24 teams from across India, accomplishing a footfall of 1000 students.
Departmental Under Graduate Committee (DUGC) Representative for 3 years .
Joint Entrance Examination, Advanced (All India Rank of 2321 , year 2021).
Joint Entrance Examination, Mains (99.24 percentile, year 2021).